



Clinical impact of adopting an AI-driven opportunistic bone health screening software using routine X-ray

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Abstract

Objectives To evaluate the clinical impact of an artificial intelligence device, Rho, that opportunistically screens X-rays for low bone mineral density (BMD; DXA T-score < −1).

Materials and methods Over 13 months, Rho analyzed X-rays from patients ≥ 50 years at a large independent health facility. Radiologists could opt to include Rho-flagged findings in their X-ray reports. DXAs that occurred within 6 months of an X-ray (from patients in the first 7 months of data collection) were categorized as being prompted by Rho (“Rho-generated”) or arranged via usual standard-of-care practice (“pre-planned”), and their outcomes (diagnoses and 10-year fracture risk scores) were compared.

Results Of 34,162 X-rays, Rho flagged 19,004 (56%) for low BMD, and radiologists included this information in 7726 (41%) reports. From the first 7 months of radiologists reporting Rho findings, initial and surveillance DXAs increased by factors of 1.8 and 1.4, respectively. Of 299 Rho-generated DXAs, 193 had low bone mass (−2.5 < T-score < −1) and 65 had osteoporosis. Rho-generated DXAs vs. pre-planned DXAs identified a greater proportion of patients with low BMD (87% vs. 69%; $p < 0.001$) and similar proportions of patients with elevated fracture risk (34% vs. 40%). The diagnostic yield was particularly marked in patients undergoing their first-ever DXA (84% vs. 60%; $p < 0.001$), and in males ≥ 65 years (83% vs. 45%; $p < 0.001$).

Conclusion Incorporating Rho in radiology workflow nearly doubled initial bone health assessments and prompted additional surveillance DXA evaluations. Rho-generated DXAs captured higher rates of true low BMD than standard-of-care practices, particularly in older men.

Keywords Osteoporosis · Radiograph · Opportunistic · Bone density · Machine learning

Abbreviations

AI	Artificial intelligence
BMD	Bone mineral density
CFRA	Clinical fracture risk assessment
DXA	Dual-energy X-ray absorptiometry
IHF	Independent health facility

Introduction

Osteoporosis is characterized by low bone mineral density (BMD) and deterioration of bone quality, and contributes to the risk of fragility fracture. Osteoporosis causes 80% of fractures in adults ≥ 50 years; 1 in 3 women and 1 in 5 men will suffer an osteoporotic fracture in their lifetime [1]. These fractures can lead to decreased quality of life and independence and excess mortality [2]. They carry a significant economic burden to healthcare systems [3, 4].

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US guidelines on osteoporosis screening recommend DXA in women aged ≥ 65 years, and in postmenopausal women aged < 65 years who are at increased risk of fracture as estimated by clinical fracture risk assessment (CFRA) [5]. In Canada, Osteoporosis Canada clinical practice guidelines recommend physicians conduct CFRA in postmenopausal females and males aged ≥ 50 years. Subsequent care, such as referral for DXA or initiation of treatment, depends on identified fracture risk factors and age [6]. In practice, CFRA are under-performed, in part due to time constraints [7], the silent nature of the disease (often the first symptom is a fracture), and the prioritization by both physicians and patients of other diseases perceived as more life threatening [7, 8]. Identification of patients with low BMD, i.e., those potentially at an early stage of osteoporosis, could prompt a CFRA and thereby enable timely intervention and management to prevent fragility fractures and related morbidity.

By visual assessment alone, it is difficult for radiologists to assess bone loss on standard X-ray until disease is pronounced [9]. In recent years, multiple research groups have demonstrated the ability of deep learning algorithms to identify patients with osteoporosis (DXA T-score ≤ -2.5) [10–14] or low BMD (DXA T-score < -1) [15–18] from conventional radiographs. Most commonly, these algorithms analyze chest X-rays [12, 13, 16, 19], though some that analyze spine [10, 14, 15] or hip/pelvis [11] X-rays have also been reported. Some predict BMD T-scores at the lumbar spine only [15] or femoral neck only [11, 16], while others predict T-scores at both [12–14, 17, 19]. Unlike other algorithms, Rho™ (16 Bit Inc., Toronto, Canada) was trained to identify low BMD from X-rays of the chest, lumbar spine, thoracic spine, pelvis, knee or hand/wrist. Rho was trained with over 60,000 X-ray/DXA pairs and has been validated in over 6600 patients, demonstrating generalizable performance across multiple datasets, X-ray body parts, sex, age, race, and technical parameters [17, 20]. By training the algorithm to predict the lowest T-score of the lumbar spine or femoral neck (regardless of whether those anatomical locations are contained in the X-ray image itself), the presence of dense structures (e.g., vascular calcifications) in the spine is less likely to mask low BMD. No manual or automated segmentation of bony structures was performed during model training, and none is required when used in clinical practice. Application in clinical practice enables “opportunistic screening,” i.e., identification of these patients when they undergo an X-ray for some other clinical indication, with no additional imaging time or radiation exposure. Rho automatically analyzes X-rays, acquired for any clinical indication, in patients aged ≥ 50 years, and sends a report to the radiologist via PACS if the patient is at risk for low BMD. The radiologist can choose to include the finding in their X-ray report, which, if included, is intended to prompt a CFRA by the referring clinician. How radiologists interact

with such an opportunistic screening device, and whether patient management is affected, has not been reported.

This study aimed to evaluate the clinical impact of an opportunistic screening device for low BMD at an Ontario-based independent health facility (IHF) with multiple locations. The rate of radiologists including Rho findings of low BMD in their X-ray reports was tracked over a 13-month period. In patients screened during the first 7 months, we compared the rate of Rho-generated and pre-planned DXAs that occurred within a 6-month time window, and compared their frequency of true low BMD, osteoporosis, and elevated 10-year fracture risk.

Materials and methods

This retrospective study utilized data from a large, multi-site ($n = 8$) IHF in Ontario, Canada. Cases screened by Rho, the rate of low BMD flagged by Rho, and the rate of radiologist reporting the Rho finding were tabulated from 2022-10-01 to 2023-10-31. To determine the rate of Rho-generated DXAs, a subset of the first 7 months was analyzed to ascertain if a subsequent DXA occurred within a 6-month time period. Rho was first licensed by Health Canada in 2022 and received de novo market authorization (2024) by the US Food and Drug Administration [20].

Once connected to the PACS, Rho screens accessions and will only analyze X-rays if (i) age ≥ 50 years, (ii) patient does not have a DXA in the PACS within the last 2 years (as if they do, their bone health is presumed to be known), and (iii) the X-ray is a frontal projection of the chest, lumbar spine, thoracic spine, pelvis, knee, or hand/wrist. If the image passes Rho’s image quality control, the image is analyzed. In Canada, the image is assigned a Rho score (1–10) that correlates with the patient’s likelihood of having low BMD (DXA T-score < -1 at the femoral neck or lumbar spine [L1–L4]). Clinical sites can decide at which Rho score to flag a patient as at risk for disease. When this threshold is met, Rho sends a report to the radiologist through PACS. In the current analyses, we considered Rho scores ≥ 6 to be “Rho Positive” and those < 6 to be “Rho Negative.”

Radiologists were made aware of the software and a potential voice dictation macro script and were instructed that it was their choice whether to include a Rho Positive finding in any of their X-ray reports. The suggested macro text was “Based on Rho, an opportunistic AI-based screening device, this patient has been identified to be at risk for having low BMD. Suggest fracture risk assessment, and if indicated, BMD assessment with DXA.” Referring clinicians were not explicitly notified of the IHF’s adoption of the software and were made aware of positive findings only if they were included in the X-ray report.

A study coordinator reviewed (i) radiologist reports of Rho Positive accessions to ascertain if the radiologist reported the Rho finding and (ii) DXA requisitions and reports for any patients who underwent a DXA following a Rho-analyzed X-ray. DXAs ordered *after* a radiologist had included a Rho Positive finding in their report were assumed to have been prompted by the Rho finding and were classified as “Rho-generated.” DXAs ordered with, or before, the Rho-analyzed X-ray, those ordered after a Rho Negative finding, or an unreported Rho Positive finding were considered “pre-planned.” The coordinator recorded responses from the clinic’s standard fracture risk questionnaire completed by patients at DXA visits with input from radiologists on the nature of the prior fracture, DXA femoral neck and L1-L4 BMD T-scores (the lower of which was used to classify patients as having true low BMD; T-score < −1), and patient’s history of DXA (self-reported in the fracture risk questionnaire or on record at the IHF). Using the fracture risk questionnaire and DXA femoral neck values, major osteoporotic FRAX scores were calculated and classified as low (< 10%), moderate (10 to < 20%), or high (≥ 20%) [1].

We classified DXAs as first-ever or surveillance DXAs, in order to ascertain rates of initial screening. Additionally, we analyzed rates of elevated fracture risk and low BMD in all DXA follow-up patients, as well as first-ever DXA follow-up patients, since conceivably patients having a repeat DXA could have a higher prevalence of risk factors (necessitating repeat DXAs) or have been on treatment that could have affected their BMD.

Given known differences in prevalence of low BMD by sex and age, analyses were conducted in subsets defined by sex and age group (50–64 years and ≥ 65 years). Comparisons between subsets or between types of DXA follow-up were analyzed with Pearson’s Chi-squared test. All records were de-identified prior to transfer to 16 Bit for analysis. Statistical analyses were performed by C.S., in R, version 4.2.2.

Results

Over 13 months, Rho screened 46,883 unique accessions from 39,356 unique patients. Rho intentionally excluded from analysis patients who had had a recent DXA in the PACS ($n = 3822$). Older (≥ 65 years) females had a higher rate of recent DXA (17%) than younger (50–64 years) females (8%), older males (7%), or younger males (0.8%; $p < 2.2\text{E-}16$, Table S1). Some X-rays failed analysis due to technical reasons ($n = 201$, 0.5%; e.g., sex not provided as female or male, instance not found), and some due to image quality ($n = 1171$; 3%). A total of 34,162 unique patients had an X-ray analyzed by Rho (Table 1), either chest, lumbar spine, thoracic spine, pelvis, knee, or hand/wrist (Table S2). A total of 19,004 patients were Rho Positive (56%) and a report was sent to PACS. Positivity rate was highest in older women and lowest in younger men. Radiologists included the Rho Positive finding in 7726 (41%) of Rho Positive patients, and the rate of reporting did not differ by sex and age group (Table 1). Rho-related activity increased over the first 6 months of the 13-month period (Figure S1). In the first month, four radiologists participated in Rho reporting, and additional radiologists were gradually recruited. By the seventh month, 14 radiologists were participating, but three opted never to report Rho findings, two of these were occasional radiologists. Restricting the reporting rate to those who reported at least one Rho finding per month, the mean (SD) monthly reporting rate was 52 (22)%, ranging from 15 to 89%. The Supplementary Material provides Rho Positivity at higher Rho score thresholds (Table S3).

DXA follow-up

To determine the rate at which Rho findings in a radiology report led to a DXA referral within 6 months of the X-ray, we restricted analysis to patients screened in the first 7 months. Rates of Rho Positive and of radiologist reporting were similar to the larger 13-month period (Table S4). Of

Table 1 Unique patients assessed by Rho over 13-month period

	Females < 65 ($N = 9293$)	Females ≥ 65 ($N = 10,627$)	Male < 65 ($N = 6685$)	Male ≥ 65 ($N = 7557$)	Total ($N = 34,162$)	p
Rho positive (% of N)	5393 (58)	8657 (82)	1728 (26)	3226 (43)	19,004 (56)	<.001
Reported (% of Rho positive)	2210 (41)	3549 (41)	702 (41)	1265 (39)	7726 (41)	0.33

Unique patients with Rho-eligible X-rays analyzed by Rho over a 13-month period, the number with low BMD identified by Rho, and the number of those that a radiologist chose to include in their report of the X-ray

the 18,519 patients with Rho-analyzed X-rays, 585 (3.2%) returned within 180 days for a pre-planned DXA (Table S4), and radiologist reporting of Rho findings led to an additional 299 Rho-generated DXAs.

Of the 585 pre-planned DXAs, 204 were the patient's first-ever and 381 were surveillance DXAs (Table S6). Radiologist reporting of Rho findings led to an additional 156 first-ever DXAs and 143 surveillance DXAs, increasing the first-ever DXA volume by a factor of 1.8 and surveillance DXA volume by a factor of 1.4. The proportion of first-ever DXAs was greater in Rho-generated vs. pre-planned DXAs (52% vs. 36%; $p < 0.001$; Table S5).

Frequency of true low BMD

Of patients that returned for DXA, ground truth BMD status by DXA could not be ascertained in 23 patients because of bilateral hip replacement ($n = 21$) or patient size exceeding DXA capacity ($n = 2$). Two of 299 Rho-generated DXAs did not have ground truth available, one due to bilateral hip replacement and one due to patient size. Of the 297 with DXA ground truth, 258 had low BMD (DXA T-score < -1): 193 had low bone mass (T-score between -2.5 and -1) and 65 had osteoporosis (T-score ≤ -2.5 ; Table S6). Rho-generated DXAs had a higher rate of low BMD than pre-planned DXAs (87% vs. 69%; Table 2). The rate of low BMD in Rho-generated DXAs did not differ by sex and age group ($\geq 83\%$; $p = 0.31$), but in pre-planned DXAs, it ranged from 45% in older males to 80% in older females ($p < 0.001$). The rates of low BMD in Rho-generated DXAs were significantly greater than those in pre-planned DXAs in each sex and age group, with the difference being most pronounced in older men (83% vs. 45%). The rates of osteoporosis in Rho-generated vs. pre-planned DXAs differed only in older males, in whom Rho-generated DXAs identified a significantly greater rate of osteoporosis (30% vs. 10%, $p = 2.2E-3$; Table S7).

Of pre-planned DXAs that were Rho Positive, 90% were true positives by DXA, and of those that were Rho Negative, 29% were false negatives; i.e., they had low BMD by DXA

(Table 2). None of the false negative patients had osteoporosis (Table S7). The rate of false negatives did not differ by sex and age group (Table 2). The positive predictive value of Rho for low BMD did not differ by X-ray body part, with values of 87%, 92%, 95%, 93%, 84%, or 87% in each of the chest ($n = 286$), lumbar spine ($n = 99$), thoracic spine ($n = 55$), pelvis ($n = 85$), knee ($n = 101$), or hand/wrist ($n = 46$); $p = 0.18$.

When restricted to a smaller group of patients who returned to have their *first-ever* DXA, Rho-generated DXAs identified 129 patients with low BMD: 99 with low bone mass and 30 with osteoporosis (Table S8). The rate of low BMD in *first-ever* Rho-generated DXAs (84%; Table S9) was similar to the rate in the larger group of Rho-generated DXAs (87%; Table 2). In contrast, the rate of *first-ever* pre-planned DXAs (60%; Table S9) was even lower than in the larger group of pre-planned DXAs (69%; $p = 0.02$). The rate of low BMD in *first-ever* Rho-generated vs. pre-planned DXAs was more than doubled in older males (78% vs. 38%), significantly greater in younger males (91% vs. 65%), and tended to be greater (but did not reach statistical significance) in younger females and older females (80% vs. 68% and 94% vs. 79%, respectively). The rates of osteoporosis in *first-ever* Rho-generated vs. pre-planned DXAs differed only in older males, in whom Rho-generated DXAs identified a significantly greater rate of osteoporosis (25% vs. 8%, $p = 0.015$; Table S10).

Frequency of elevated fracture risk

When restricted to patients who returned to have their *first-ever* DXA, Rho-generated DXAs had a similar rate of elevated fracture risk as pre-planned DXA (24% and 26%, respectively; Table 3). Rho-generated DXAs had higher rates of elevated fracture risk (24%) than pre-planned DXAs that followed a Rho Negative finding (3%), but lower rates than pre-planned DXAs that followed a Rho Positive finding (44%; Table 3). Results were similar when considering all follow-up DXAs (Table S11). Results for fracture risk

Table 2 Rate of low BMD in DXA follow-up patients

	Rho-generated DXA	Pre-planned DXA			DXA comparison	
		All	Rho Negative	Rho Positive	p^a	p^b
Female 50–64 (%)	91/108 (84)	98/141 (70)	13/51 (26)	85/90 (94)	0.007	<.001
Female ≥ 65 (%)	96/106 (91)	225/281 (80)	21/52 (40)	204/229 (89)	0.014	<.001
Male 50–64 (%)	24/26 (92)	18/31 (58)	4/16 (25)	14/15 (93)	0.003	<.001
Male ≥ 65 (%)	47/57 (83)	50/111 (45)	16/70 (23)	34/41 (83)	<.001	<.001
Total (%)	258/297 (87)	391/564 (69)	54/189 (29)	337/375 (90)	2.2E-8	<.001
p between sex and age group	0.31	<.001	0.17	0.20		

^a p -value between Rho-generated DXAs vs. all pre-planned DXAs

^b p -value between Rho-generated DXAs, pre-planned Rho Positive DXAs and pre-planned Rho Negative DXAs

Table 3 Elevated fracture risk in first-ever DXA follow-up patients

	Rho-generated DXA	Pre-planned DXA			DXA comparison	
		All	Rho Negative	Rho Positive	p^a	p^b
Female 50–64 (%)	8/63 (13)	9/61 (15)	1/24 (4)	8/37 (22)	0.74	0.15
Female ≥ 65 (%)	17/31 (55)	25/34 (74)	1/4 (25)	24/30 (80)	0.12	0.028
Male 50–64 (%)	2/23 (9)	3/20 (15)	0/9 (0)	3/11 (27)	0.52	0.14
Male ≥ 65 (%)	9/35 (26)	7/54 (13)	0/36 (0)	7/18 (39)	0.13	6.6E-4
Total (%)	36/152 (24)	44/169 (26)	2/73 (2.7)	42/96 (44)	0.63	7.2E-9
p between sex and age group	<.001	<.001	0.031	<.001		

^a p -value between Rho-generated DXAs vs. all pre-planned DXAs

^b p -value between Rho-generated DXAs, pre-planned Rho Positive DXAs, and pre-planned Rho Negative DXAs

rates when further split into low, moderate, and high categories for each sex and age group and at different Rho score thresholds are presented in Tables S12–S15 and S16–S19, respectively.

Discussion

Adoption of Rho at the IHF helped prompt DXA referrals in patients found to have low BMD and elevated fracture risk, many of whom were previously undiagnosed. Radiologists including Rho findings in their X-ray reports generated increased both *initial* bone health assessments and surveillance DXAs. These Rho-generated DXAs had a significantly higher diagnostic yield than pre-planned DXA as they demonstrated a higher rate of low BMD by DXA. Rho-generated DXA vs. pre-planned DXAs had an even more pronounced diagnostic yield in patients who returned for a first-ever DXA (Table S5) and in older men (Tables 2 and S7). While rates of elevated fracture risk were similar between Rho-generated DXAs and pre-planned DXAs (Table 3), in older men Rho-generated DXA vs. pre-planned DXAs tended to identify higher rates of elevated fracture risk. CFRAs are not always a priority in patients, and osteoporosis evaluation is low, particularly in older men [21]. These findings suggest that a Rho finding triggered an appropriate CFRA and referral for DXA, and conceivably, facilitated timely (earlier) treatment and prevention strategies.

While an increase in CFRAs will add to physician workload, Rho is prompting this relatively short and simple assessment, *targeting* patients who already have at least one risk factor (low BMD on X-ray). Feedback from referring providers was not collected, but the IHF was unaware of any patient or provider complaints associated with adoption of the software. Alternatives to primary care follow-up could be explored as a means of reducing provider workload, e.g., adding *primary* fracture prevention to fracture liaison services. Triggering CFRAs and subsequent DXAs will also incur healthcare system costs but may ultimately save

costs through reducing fracture-associated costs. In the UK, the SCOOP trial demonstrated that conducting CFRAs in women ≥ 70 years saved 9 hip fractures and 20 non-hip fractures/thousand patients over the remaining lifetime (mean 14 years), saved £286 per patient, and gained 0.015 quality-adjusted life years (QALYs) per patient compared with usual management [22]. In Canada and the USA, fractures and associated costs (e.g., hospitalization and long-term disability) are estimated at over \$1.9 billion/year [23] and \$57 billion/year [4], respectively. Country-specific economic and resource modeling should be conducted to determine whether the potential to reduce fractures and associated costs outweighs the costs of *targeted* CFRAs and downstream management (e.g., DXA referrals, pharmacotherapy).

In this IHF, some radiologists chose never to include Rho findings in their X-ray reports. Of those who included at least one Rho finding per month, the rate of inclusion ranged widely: the mean reporting rate was 52% and did not differ by patient sex or age group, and the range was 15 to 89%. To maximize the potential benefits of opportunistic X-ray screening requires both radiologists and referring clinicians to be aware of its efficacy. In both types of care providers, awareness of (i) the software performance and intended use, (ii) the osteoporosis care gap, and (iii) the availability of effective treatments could increase the clinical utility of opportunistic screening. One additional means of maximizing conveyance of Rho findings to referring clinicians could be to automate reporting of Rho-positive findings. While some radiologists might appreciate this as a reduction in their workload, others might object to automatic inclusion as they may prefer to appropriately exclude the findings in some patients (e.g., in patients with low expected lifespan due to terminal illness).

Of patients with Rho Positive findings reported by the radiologist, 7% returned for a Rho-generated DXA. Multiple factors affect this rate of Rho-generated DXAs, most notably (i) that referring providers received no formal education about the brand-new technology and (ii) the 6-month observation period. Although DXA appointment wait times

were much less than 6 months, many providers and patients may not have acted in response to the Rho finding within the 6-month period. Some referring providers may not have acted on the finding at all; others might already have been aware of the patient's bone health status from a prior DXA that occurred outside the IHF, and it is probable that some patients underwent follow-up DXA at a different imaging facility. Alternatively, some providers might have conducted a CFRA and determined a DXA was not appropriate or needed at this time. Importantly, this step of conducting a CFRA, even for patients identified by FRAX as at low or moderate risk, presents an opportunity for care providers and patients to discuss lifestyle measures to optimize bone health and reduce fracture risk.

Finally, while operating at the manufacturer's default threshold (Rho score ≥ 6), the rates of Rho-flagged low BMD in younger and older females (58% and 82%) and males (26% and 43%, respectively; Table 1) were similar to rates of low BMD in the general population [24]. The true positive rate for low BMD in Rho Positive patients was high (87% in Rho-generated DXAs and 90% in pre-planned DXAs that followed an unreported Rho Positive). The true positive rate did not differ by type of X-ray, consistent with previous publications that demonstrated consistency of the algorithm across X-ray body parts [17, 20]. The rate of true low BMD in Rho Negative patients (i.e., false negatives) was low (29%), similar to published performance in other datasets [17]. As Rho Negative findings are not reported, false negative patients still receive the current standard of care. Multiple other algorithms have demonstrated similar performance at predicting low BMD from standard chest, or pelvis/hip X-rays as part of their algorithm validation [15, 16, 18, 19]. To the best of our knowledge, they have not published results using prospective datasets acquired through incorporation of a device into routine clinical practice.

Strengths of the study include the real-world application of Rho in multiple locations of a large IHF and the ability to track follow-up DXA results. However, as described in the methods, classification of DXA follow-ups as Rho-generated relied on the assumption that a reported Rho Positive finding had prompted the referral. Radiologists were not provided with any educational or promotional material about Rho by design, and the lack of awareness of evidence supporting Rho's clinical efficacy may have lowered their adoption rates. Collecting feedback from radiologists and referring clinicians could have provided insight on clinician attitudes of opportunistic screens such as Rho. No statistically significant difference in diagnostic performance was observed by X-ray type, but continued validation in real-world clinical use is warranted to understand the effects of injury or degeneration. An additional limitation was that a patient's history of DXA relied on their memory and/or the DXA being on record at the IHF.

In both men and women, implementation of an AI-driven opportunistic screening software for low BMD prompted DXA referrals in patients found to have low BMD and elevated fracture risk, many of whom were previously undiagnosed. Men in particular benefited from AI-driven opportunistic screening, as Rho-generated DXAs had higher diagnostic yield (higher rates of low BMD and osteoporosis) than DXAs ordered by the current standard of care. Given that X-rays are widely performed, through adoption of AI-driven opportunistic screening software, radiologists can help address the care gap in osteoporosis by providing a "smart nudge" to referring physicians to conduct the often-overlooked CFRA. Future research directions include whether integration of Rho into osteoporosis screening pathways can motivate patients with early signs of disease to commit to bone health optimization strategies, including lifestyle changes (e.g., smoking and alcohol cessation, weight-bearing exercise) or pharmacotherapy compliance, whether Rho can improve fracture risk estimates when DXA is not available, and, ultimately, whether its inclusion in clinical care reduces incident fragility fractures.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1007/s00256-026-05148-9>.

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Data availability The data for this study is not available.

Declarations

Ethics approval All procedures performed in studies involving human participants were in accordance with the ethical standards of an independent review board (Veritas, Montreal, Canada) and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent Approval from the independent review board was obtained and in keeping with the policies for a retrospective review, informed consent was not required.

Conflict of interest CS reports a relationship with 16 Bit Inc. that includes employment and equity or stocks. ML reports a relationship with 16 Bit Inc. that includes employment. AB reports a relationship with 16 Bit Inc. that includes equity or stocks and a patent for Systems and Methods for Approximating Bone Mineral Density and Fracture Risk Using Single Energy X-Rays pending to 16 Bit. MC reports a relationship with 16 Bit Inc. that includes equity or stocks, and a patent for Systems and Methods for Approximating Bone Mineral Density and Fracture Risk Using Single Energy X-Rays pending to 16 Bit.

References

1. Kanis JA on behalf of the World Health Organization Scientific Group. Assessment of osteoporosis at the primary health-care level. Technical Report. World Health Organization Collaborating

- Centre for Metabolic Bone Diseases, University of Sheffield. UK: Printed by the University of Sheffield. 2007.
2. Fc GBD. Global, regional, and national burden of bone fractures in 204 countries and territories, 1990–2019: a systematic analysis from the Global Burden of Disease Study 2019. *Lancet Healthy Longev.* 2021;2:e580–92.
 3. Hopkins RB, Burke N, Von Keyserlingk C, et al. The current economic burden of illness of osteoporosis in Canada. *Osteoporos Int.* 2016;27:3023–32.
 4. Lewiecki EM, Ortendahl JD, Vanderpuye-Orgle J, et al. Health-care policy changes in osteoporosis can improve outcomes and reduce costs in the United States. *JBM R Plus.* 2019;3:e10192.
 5. US Preventive Services Task Force, Nicholson WK, Silverstein M et al. Screening for osteoporosis to prevent fractures: US Preventive Services Task Force recommendation statement. *JAMA.* 2025;333(6):498–508
 6. Morin SN, Feldman S, Funnell L, et al. Clinical practice guideline for management of osteoporosis and fracture prevention in Canada: 2023 update. *CMAJ.* 2023;195:E1333–48.
 7. Singer AJ, Sharma A, Deignan C, Borgermans L. Closing the gap in osteoporosis management: the critical role of primary care in bone health. *Curr Med Res Opin.* 2023;39:387–98.
 8. Kendler DL, Adachi JD, Brown JP, et al. A scorecard for osteoporosis in Canada and seven Canadian provinces. *Osteoporos Int.* 2021;32:123–32.
 9. Brunader R, Shelton DK. Radiologic bone assessment in the evaluation of osteoporosis. *Am Fam Physician.* 2002;65:1357–64.
 10. Hsieh CI, Zheng K, Lin C, et al. Automated bone mineral density prediction and fracture risk assessment using plain radiographs via deep learning. *Nat Commun.* 2021;12:5472.
 11. Jang R, Choi JH, Kim N, Chang JS, Yoon PW, Kim CH. Prediction of osteoporosis from simple hip radiography using deep learning algorithm. *Sci Rep.* 2021;11:19997.
 12. Tsai DJ, Lin C, Lin CS, Lee CC, Wang CH, Fang WH. Artificial intelligence-enabled chest x-ray classifies osteoporosis and identifies mortality risk. *J Med Syst.* 2024;48:12.
 13. Lin C, Tsai DJ, Wang CC, et al. Osteoporotic precise screening using chest radiography and artificial neural network: the OPSCAN randomized controlled trial. *Radiology.* 2024;311:e231937.
 14. Hong N, Cho SW, Lee YH, et al. Deep learning-based identification of vertebral fracture and osteoporosis in lateral spine radiographs and DXA vertebral fracture assessment to predict incident fracture. *J Bone Miner Res.* 2025;40:628–38.
 15. Zhang B, Yu K, Ning Z, et al. Deep learning of lumbar spine X-ray for osteopenia and osteoporosis screening: a multicenter retrospective cohort study. *Bone.* 2020;140:115561.
 16. Asamoto T, Takegami Y, Sato Y, et al. External validation of a deep learning model for predicting bone mineral density on chest radiographs. *Arch Osteoporos.* 2024;19:15.
 17. Bilbily A, Syme CA, Adachi JD, et al. Opportunistic screening of low bone mineral density from standard X-rays. *J Am Coll Radiol.* 2024;21:633–9.
 18. Pignolo RJ, Connell JJ, Briggs W, et al. Opportunistic assessment of osteoporosis using hip and pelvic X-rays with OsteoSight™: validation of an AI-based tool in a US population. *Osteoporos Int.* 2025;36:1053–60.
 19. Jang M, Kim M, Bae SJ, Lee SH, Koh JM, Kim N. Opportunistic osteoporosis screening using chest radiographs with deep learning: development and external validation with a cohort dataset. *J Bone Miner Res.* 2022;37:369–77.
 20. US FDA. De Novo classification request for Rho. 2025. https://www.accessdata.fda.gov/cdrh_docs/reviews/DEN230023.pdf. Accessed 5 Nov 2025.
 21. Narla RR, Hirano LA, Lo SHY, Anawalt BD, Phelan EA, Matsumoto AM. Suboptimal osteoporosis evaluation and treatment in older men with and without additional high-risk factors for fractures. *J Investig Med.* 2019;67:743–9.
 22. Söreskog E, Borgström F, Shepstone L, et al. Long-term cost-effectiveness of screening for fracture risk in a UK primary care setting: the SCOOP study. *Osteoporos Int.* 2020;31:1499–506.
 23. Tarride JÉ, Adachi JD, Brown JP, Schemitsch E, Slatkovska L, Burke N. Incremental costs of fragility fractures: a population-based matched -cohort study from Ontario, Canada. *Osteoporos Int.* 2021;32:1753–61.
 24. Wright NC, Looker AC, Saag KG, et al. The recent prevalence of osteoporosis and low bone mass in the United States based on bone mineral density at the femoral neck or lumbar spine. *J Bone Miner Res.* 2014;29:2520–6.

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